

```
import pandas as pd
```

```
df=pd.read_excel("/content/updated.xlsx")
```

```
df
```

	student_id	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent
0	S001	kamalika	20	female	urban	group B	standard	8.0	8.8	72.1
1	S002	swathi	12	female	rural	group C	standard	1.3	8.6	60.7
2	S003	abinaya	21	female	urban	group B	standard	4.0	8.2	73.7
3	S004	selvam	23	male	urban	group A	free/reduced	3.5	4.8	95.1
4	S005	saravanan	24	male	rural	group C	standard	9.1	6.4	89.8
5	S006	harini	23	female	urban	group B	standard	8.4	5.1	58.5
6	S007	harshini	24	female	rural	group B	standard	10.8	6.0	54.2
7	S008	adithya	20	male	rural	group B	free/reduced	2.0	4.3	75.8
8	S009	vasi	23	male	rural	group D	free/reduced	5.6	5.9	81.6
9	S010	ruthvika	21	female	rural	group B	free/reduced	1.3	8.9	66.8
10	S011	prajan	23	male	urban	group C	standard	3.4	5.3	90.9
11	S012	krithick	24	male	urban	group D	standard	6.6	7.9	87.6
12	S013	deshika	21	female	urban	group B	standard	1.3	6.3	83.6
13	S014	roshan	23	male	rural	group A	standard	3.2	6.1	61.2
14	S015	sneha	24	female	urban	group A	standard	8.1	8.8	60.0
15	S016	priya	21	female	rural	group C	standard	7.0	9.0	51.2
16	S017	jeeva	23	male	rural	group C	standard	3.4	6.8	62.2
17	S018	nive	22	female	rural	group B	free/reduced	7.5	7.6	73.8
18	S019	iyappan	24	male	urban	group C	free/reduced	9.9	4.8	92.5
19	S020	jaya shree	21	female	urban	group C	free/reduced	1.1	5.5	53.6
20	S021	karthi	20	male	urban	group D	standard	9.9	8.8	70.7
21	S022	boomika	24	female	rural	group B	free/reduced	8.7	6.9	81.5
22	S023	lokesh	23	male	rural	group D	standard	4.7	6.7	59.7

```
print(df.head())
```

	student_id	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent	Unnamed: 10	reading score	math score	writing score	previous_scores	exam_score	result
0	S001	kamalika	20	female	urban	group B	standard	8.0	8.8	72.1	online	72	72	74	45	30.2	Fail
1	S002	swathi	12	female	rural	group C	standard	1.3	8.6	60.7	yes	90	69	88	55	25.0	Fail
2	S003	abinaya	21	female	urban	group B	standard	4.0	8.2	73.7	no	95	90	93	86	35.8	Pass
3	S004	selvam	23	male	urban	group A	free/reduced	3.5	4.8	95.1	yes	57	47	44	66	34.0	Fail
4	S005	saravanan	24	male	rural	group C	standard	9.1	6.4	89.8	yes	78	76	75	71	40.3	Pass

```
print(df.tail())
```

	student_id	name	age	gender	address	race/ethnicity	lunch
18	S019	iyappan	24	male	urban	group C	free/reduced
19	S020	jaya shree	21	female	urban	group C	free/reduced
20	S021	karthi	20	male	urban	group D	standard
21	S022	boomika	24	female	rural	group B	free/reduced
22	S023	lokesh	23	male	rural	group D	standard

	hours_studied	sleep_hours	attendance_percent	Unnamed: 10	reading score	\
18	9.9	4.8	92.5	no	42	
19	1.1	5.5	53.6	yes	58	
20	9.9	8.8	70.7	yes	69	
21	8.7	6.9	81.5	yes	75	
22	4.7	6.7	59.7	yes	54	

	math score	writing score	previous_scores	exam_score	result
18	46	46	54	35.6	Pass
19	54	61	65	17.1	Fail
20	66	63	84	46.0	Pass
21	65	70	55	36.1	Pass
22	44	53	59	29.6	Fail

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23 entries, 0 to 22
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   student_id            23 non-null    object
1   name                  23 non-null    object
2   age                   23 non-null    int64
3   gender                23 non-null    object
4   address               23 non-null    object
5   race/ethnicity        23 non-null    object
6   lunch                 23 non-null    object
7   hours_studied         23 non-null    float64
8   sleep_hours           23 non-null    float64
9   attendance_percent     23 non-null    float64
10  Unnamed: 10           23 non-null    object
11  reading score         23 non-null    int64
12  math score            23 non-null    int64
13  writing score          23 non-null    int64
14  previous_scores       23 non-null    int64
15  exam_score            23 non-null    float64
16  result                23 non-null    object
dtypes: float64(4), int64(5), object(8)
memory usage: 3.2+ KB
None
```

```
print(df.describe())
```

	age	hours_studied	sleep_hours	attendance_percent	\
count	23.000000	23.00000	23.000000	23.000000	
mean	21.913043	5.60000	6.847826	72.034783	
std	2.592174	3.19232	1.532932	13.723689	
min	12.000000	1.10000	4.300000	51.200000	
25%	21.000000	3.30000	5.700000	60.350000	
50%	23.000000	5.60000	6.700000	72.100000	
75%	23.500000	8.25000	8.400000	82.600000	
max	24.000000	10.80000	9.000000	95.100000	

	reading score	math score	writing score	previous_scores	exam_score
count	23.000000	23.000000	23.000000	23.000000	23.000000
mean	67.086957	60.695652	64.391304	67.434783	32.486957
std	17.528971	18.309051	17.900582	14.963064	6.810841
min	32.000000	18.000000	28.000000	41.000000	17.100000
25%	54.000000	46.500000	51.000000	55.000000	29.450000
50%	69.000000	65.000000	67.000000	68.000000	34.100000
75%	79.500000	71.500000	76.500000	82.500000	35.950000
max	95.000000	90.000000	93.000000	90.000000	46.000000

```
print(df.shape)
```

```
(23, 17)
```

```
print(df.columns)
```

```
Index(['student_id', 'name', 'age', 'gender', 'address', 'race/ethnicity',
      'lunch', 'hours_studied', 'sleep_hours', 'attendance_percent',
      'Unnamed: 10', 'reading score', 'math score', 'writing score',
      'previous_scores', 'exam_score', 'result'],
      dtype='object')
```

```
df["name"].str.replace("iyappan","jeeva")
```

	name
0	kamalika
1	swathi
2	abinaya
3	selvam
4	saravanan
5	harini
6	harshini
7	adithya
8	vasi
9	ruthvika
10	prajan
11	krithick
12	deshika
13	roshan
14	sneha
15	priya
16	jeeva
17	nive
18	jeeva
19	jaya shree
20	karthi
21	boomika
22	lokesh

dtype: object

```
df["name"].str.split(" ")
```

	name
0	[kamalika]
1	[swathi]
2	[abinaya]
3	[selvam]
4	[saravanan]
5	[harini]
6	[harshini]
7	[adithya]
8	[vasi]
9	[ruthvika]
10	[prajan]
11	[krithick]
12	[deshika]
13	[roshan]
14	[sneha]
15	[priya]
16	[jeeva]
17	[nive]
18	[iyappan]
19	[jaya, shree]
20	[karthi]
21	[boomika]
22	[lokesh]

dtype: object

```
df["name"].str.startswith("s")
```

	name
--	------

0	False
---	-------

1	True
---	------

2	False
---	-------

3	True
---	------

4	True
---	------

5	False
---	-------

6	False
---	-------

7	False
---	-------

8	False
---	-------

9	False
---	-------

10	False
----	-------

11	False
----	-------

12	False
----	-------

13	False
----	-------

14	True
----	------

15	False
----	-------

16	False
----	-------

17	False
----	-------

18	False
----	-------

19	False
----	-------

20	False
----	-------

21	False
----	-------

22	False
----	-------

dtype: bool

```
df["name"].str.upper()
```

	name
0	KAMALIKA
1	SWATHI
2	ABINAYA
3	SELVAM
4	SARAVANAN
5	HARINI
6	HARSHINI
7	ADITHYA
8	VASI
9	RUTHVIKA
10	PRAJAN
11	KRITHICK
12	DESHIKA
13	ROSHAN
14	SNEHA
15	PRIYA
16	JEEVA
17	NIVE
18	IYAPPAN
19	JAYA SHREE
20	KARTHI
21	BOOMIKA
22	LOKESH

dtype: object

```
df["name"].str.lower()
```

	name
0	kamalika
1	swathi
2	abinaya
3	selvam
4	saravanan
5	harini
6	harshini
7	adithya
8	vasi
9	ruthvika
10	prajan
11	krithick
12	deshika
13	roshan
14	sneha
15	priya
16	jeeva
17	nive
18	iyappan
19	jaya shree
20	karthi
21	boomika
22	lokesh

dtype: object

```
df["name"].str.capitalize()
```

name

0 Kamalika
1 Swathi
2 Abinaya
3 Selvam
4 Saravanan
5 Harini
6 Harshini
7 Adithya
8 Vasi
9 Ruthvika
10 Prajan
11 Krithick
12 Deshika
13 Roshan
14 Sneha
15 Priya
16 Jeeva
17 Nive
18 lyappan
19 Jaya shree
20 Karthi
21 Boomika
22 Lokesh

dtype: object

df[(df["age"]<18)]

	student_id	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent	Unnamed: 10
1	S002	swathi	12	female	rural	group C	standard	1.3	8.6	60.7	yes

df[df["name"].str.startswith("h")]

	student_id	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent	Unnamed: 10
5	S006	harini	23	female	urban	group B	standard	8.4	5.1	58.5	y
6	S007	harshini	24	female	rural	group B	standard	10.8	6.0	54.2	

df[df["name"].str.endswith("i")]

	student_id	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent	Un
1	S002	swathi	12	female	rural	group C	standard	1.3	8.6	60.7	
5	S006	harini	23	female	urban	group B	standard	8.4	5.1	58.5	
6	S007	harshini	24	female	rural	group B	standard	10.8	6.0	54.2	
8	S009	vasi	23	male	rural	group D	free/reduced	5.6	5.9	81.6	
20	S021	karthi	20	male	urban	group D	standard	9.9	8.8	70.7	

df["reading_score"].mean()

np.float64(67.08695652173913)


```
df["reading_score"].sum()
```

```
np.int64(1543)
```

```
df1=pd.read_excel("/content/new.xlsx")
```

df1

	student_id	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent
0	S001	kamalika	20	female	urban	group B	standard	8.0	8.8	72.1
1	S002	swathi	12	female	rural	group C	standard	1.3	8.6	60.7
2	S003	abinaya	21	female	urban	group B	standard	4.0	8.2	73.7
3	S004	selvam	23	male	urban	group A	free/reduced	3.5	4.8	95.1
4	S005	saravanan	24	male	rural	group C	standard	9.1	6.4	89.8
5	S006	harini	23	female	urban	group B	standard	8.4	5.1	58.5
6	S007	harshini	24	female	rural	group B	standard	10.8	6.0	54.2
7	S008	adithya	20	male	rural	group B	free/reduced	2.0	4.3	75.8
8	S009	vasi	23	male	rural	group D	free/reduced	5.6	5.9	81.6
9	S010	ruthvika	21	female	rural	group B	free/reduced	1.3	8.9	66.8
10	S011	prajan	23	male	urban	group C	standard	3.4	5.3	90.9
11	S012	krithick	24	male	urban	group D	standard	6.6	7.9	87.6
12	S013	deshika	21	female	urban	group B	standard	1.3	6.3	83.6
13	S014	roshan	23	male	rural	group A	standard	3.2	6.1	61.2
14	S015	sneha	24	female	urban	group A	standard	8.1	8.8	60.0
15	S016	priya	21	female	rural	group C	standard	7.0	9.0	51.2
16	S017	jeeva	23	male	rural	group C	standard	3.4	6.8	62.2
17	S018	nive	22	female	rural	group B	free/reduced	7.5	7.6	73.8
18	S019	iyappan	24	male	urban	group C	free/reduced	9.9	4.8	92.5
19	S020	jaya shree	21	female	urban	group C	free/reduced	1.1	5.5	53.6
20	S021	karthi	20	male	urban	group D	standard	9.9	8.8	70.7
21	S022	boomika	24	female	rural	group B	free/reduced	8.7	6.9	81.5
22	S023	lokesh	23	male	rural	group D	standard	4.7	6.7	59.7

```
df["hours_studied"].astype(int)
```

hours_studied

0	8
1	1
2	4
3	3
4	9
5	8
6	10
7	2
8	5
9	1
10	3
11	6
12	1
13	3
14	8
15	7
16	3
17	7
18	9
19	1
20	9
21	8
22	4

dtype: int64

```
df["name"]  
df[["name", "hours_studied"]]
```

	name	hours_studied
0	kamalika	8.0
1	swathi	1.3
2	abinaya	4.0
3	selvam	3.5
4	saravanan	9.1
5	harini	8.4
6	harshini	10.8
7	adithya	2.0
8	vasi	5.6
9	ruthvika	1.3
10	prajan	3.4
11	krithick	6.6
12	deshika	1.3
13	roshan	3.2
14	sneha	8.1
15	priya	7.0
16	jeeva	3.4
17	nive	7.5
18	iyappan	9.9
19	jaya shree	1.1
20	karthi	9.9
21	boomika	8.7
22	lokesh	4.7

```
df.drop("student_id", axis=1)
```

	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent	Unnamed: 10	re
0	kamalika	20	female	urban	group B	standard	8.0	8.8	72.1		online
1	swathi	12	female	rural	group C	standard	1.3	8.6	60.7		yes
2	abinaya	21	female	urban	group B	standard	4.0	8.2	73.7		no
3	selvam	23	male	urban	group A	free/reduced	3.5	4.8	95.1		yes
4	saravanan	24	male	rural	group C	standard	9.1	6.4	89.8		yes
5	harini	23	female	urban	group B	standard	8.4	5.1	58.5		yes
6	harshini	24	female	rural	group B	standard	10.8	6.0	54.2		no
7	adithya	20	male	rural	group B	free/reduced	2.0	4.3	75.8		no
8	vasi	23	male	rural	group D	free/reduced	5.6	5.9	81.6		no
9	ruthvika	21	female	rural	group B	free/reduced	1.3	8.9	66.8		yes
10	prajan	23	male	urban	group C	standard	3.4	5.3	90.9		no
11	krithick	24	male	urban	group D	standard	6.6	7.9	87.6		yes
12	deshika	21	female	urban	group B	standard	1.3	6.3	83.6		no
13	roshan	23	male	rural	group A	standard	3.2	6.1	61.2		no
14	sneha	24	female	urban	group A	standard	8.1	8.8	60.0		yes
15	priya	21	female	rural	group C	standard	7.0	9.0	51.2		yes
16	jeeva	23	male	rural	group C	standard	3.4	6.8	62.2		yes
17	nive	22	female	rural	group B	free/reduced	7.5	7.6	73.8		no
18	iyappan	24	male	urban	group C	free/reduced	9.9	4.8	92.5		no
19	jaya shree	21	female	urban	group C	free/reduced	1.1	5.5	53.6		yes
20	karthi	20	male	urban	group D	standard	9.9	8.8	70.7		yes
21	boomika	24	female	rural	group B	free/reduced	8.7	6.9	81.5		yes
22	lokesh	23	male	rural	group D	standard	4.7	6.7	59.7		yes

```
df.drop(0, axis=0)
```

	student_id	name	age	gender	address	race/ethnicity	lunch	hours_studied	sleep_hours	attendance_percent
1	S002	swathi	12	female	rural	group C	standard	1.3	8.6	60.7
df.fillna(0)										
3	S004	selvam	23	male	urban	group A	free/reduced	3.5	4.8	95.1
4	S005	saravanan	24	male	rural	group C	standard	9.1	6.4	89.8
0	S001	kamalika	20	female	urban	group B	standard	8.0	8.8	72.1
1	S002	swathi	12	female	rural	group C	standard	1.3	8.6	60.7
6	S007	harshini	24	female	rural	group B	standard	10.8	6.0	54.2
2	S003	abinaya	21	female	urban	group B	standard	4.0	8.2	73.7
3	S004	selvam	23	male	urban	group A	free/reduced	3.5	4.8	95.1
4	S005	saravanan	24	male	rural	group C	standard	9.1	6.4	89.8
5	S006	harini	23	female	urban	group B	standard	8.4	5.1	58.5
10	S011	prajan	23	male	urban	group C	standard	3.4	5.3	90.9
6	S007	harshini	24	female	rural	group B	standard	10.8	6.0	54.2
7	S008	adithya	20	male	rural	group B	free/reduced	2.0	4.3	75.8
12	S013	deshika	21	female	urban	group B	standard	1.3	6.3	83.6
8	S009	vasi	23	male	rural	group D	free/reduced	5.6	5.9	81.6
9	S010	ruthvika	21	female	rural	group B	free/reduced	1.3	8.8	66.8
14	S015	sneha	24	female	urban	group A	standard	8.1	8.8	60.0
10	S011	prajan	23	male	urban	group C	standard	3.4	5.3	90.9
11	S012	krithick	24	male	urban	group B	standard	6.6	7.9	87.6
16	S017	jeeva	23	male	rural	group C	standard	3.4	6.8	62.2
12	S013	deshika	21	female	urban	group B	standard	1.3	6.3	83.6
13	S014	roshan	23	male	rural	group A	free/reduced	3.2	6.1	61.2
18	S019	iyappan	24	male	urban	group C	standard	9.9	8.8	70.7
14	S015	sneha	24	female	urban	group A	standard	8.1	8.8	60.0
15	S016	priya	21	female	rural	group C	standard	7.0	9.0	51.2
20	S021	karthi	20	female	urban	group D	standard	9.9	8.8	70.7
16	S017	jeeva	23	male	rural	group C	standard	3.4	6.8	62.2